LIVESTOCK GUARDIAN

User Manual

Version 1.0

Project Title: Livestock Guardian – AI-Powered Livestock Identification and Theft Prevention System

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Technology Stack: Android (Java), FastAPI, Supabase, ONNX AI Model, Render Cloud

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1. Introduction

Livestock Guardian is an Artificial Intelligence-powered Android application designed to assist livestock owners, farmers, and agricultural stakeholders in managing and protecting their animals through biometric identification.

Traditional livestock identification methods such as ear tags and paper records can be lost, damaged, or manipulated. Livestock Guardian uses muzzle biometric recognition to create a unique digital identity for each animal.

The system enables farmers to:

- Register livestock
- Identify animals using muzzle biometrics
- Verify ownership
- Report theft incidents
- Track ownership history
- Maintain centralized digital livestock records

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2. Project Overview

The application combines:

- Android Mobile Application
- AI-Based Muzzle Recognition
- Cloud Database Management
- Theft Reporting System
- Ownership Verification

The platform provides a secure and efficient method for managing livestock information while reducing ownership disputes and livestock theft.

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3. System Features

User Management

- User Registration
- Secure Login
- Session Management
- User Profile Management

Livestock Management

- Animal Registration
- Animal Profile Creation
- Animal Information Storage
- Livestock History Tracking

AI Features

- Muzzle Biometric Enrollment
- Duplicate Animal Detection
- Animal Identification
- AI-Based Matching

Theft Prevention

- Theft Reporting
- Stolen Animal Tracking
- Ownership Verification

Cloud Features

- Real-Time Data Storage
- Centralized Database
- Cloud Synchronization

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4. Technology Stack

Mobile Application

- Java
- Android Studio
- Material Design 3
- Retrofit
- OkHttp

Backend

- FastAPI
- Python

Artificial Intelligence

- ONNX Model
- Deep Learning
- Biometric Recognition

Database

- Supabase PostgreSQL
- Supabase Authentication
- Supabase Storage

Deployment

- Render Cloud Platform

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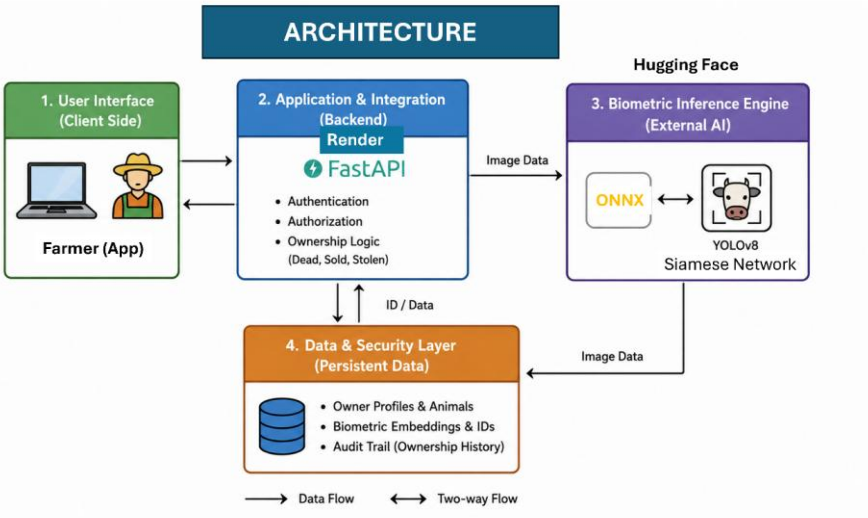
5. System Architecture

The application follows a client-server architecture.

Flow



Components

1. Android Application
2. FastAPI Server
3. AI Recognition Engine
4. Supabase Database
5. Cloud Storage Services

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6. Installation Guide

Requirements

- Android 8.0 or Higher
- Internet Connection
- Minimum 3 GB RAM
- Camera Access

Installation Steps

Step 1

Download the APK file.

Step 2

Enable installation from unknown sources.

Step 3

Open the APK file.

Step 4

Install the application.

Step 5

Launch Livestock Guardian.

Step 6

Register a new account or login.

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7. User Registration

The registration feature allows new users to create an account.

Steps

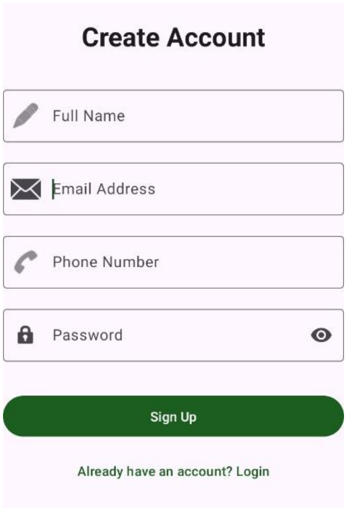
1. Open the application.
2. Tap Register.
3. Enter Full Name.
4. Enter Email Address.
5. Enter Phone Number.
6. Create Password.

7. Submit Registration Form.

Expected Result

User account is created successfully and stored securely.

Screenshot



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8. User Login

Users can access the application through secure authentication.

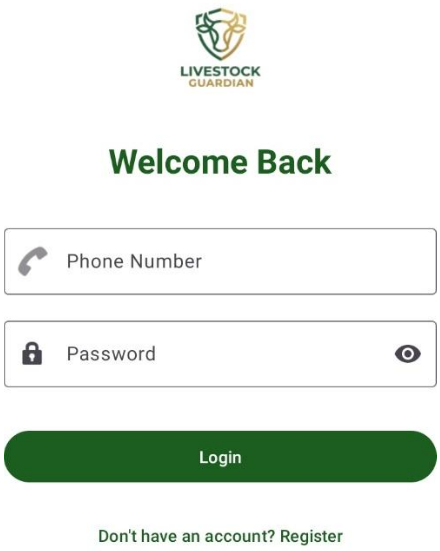
Steps

1. Open Application.
2. Enter Email Address.
3. Enter Password.
4. Tap Login.

Expected Result

User is redirected to Dashboard.

Screenshot



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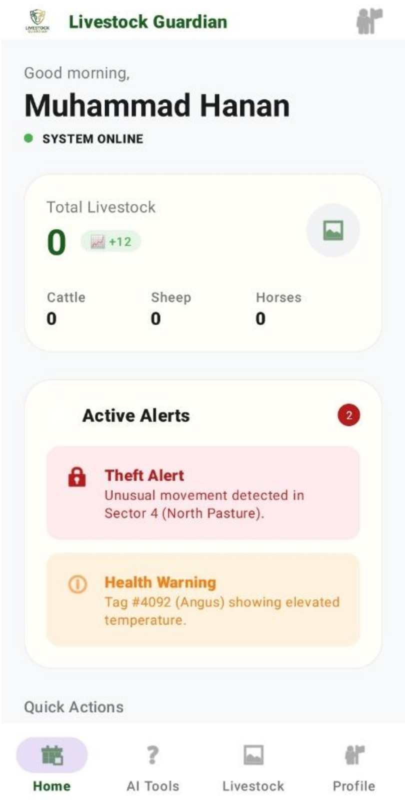
9. Dashboard

The dashboard provides a summary of livestock information.

Dashboard Features

- Total Animals
- Registered Livestock
- Theft Reports
- Quick Actions
- Navigation Menu

Screenshot



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10. Livestock Registration

This module registers a new animal in the system.

Required Information

- Animal Name
- Species
- Breed
- Gender
- Age
- Weight
- Color
- Additional Notes

Biometric Enrollment

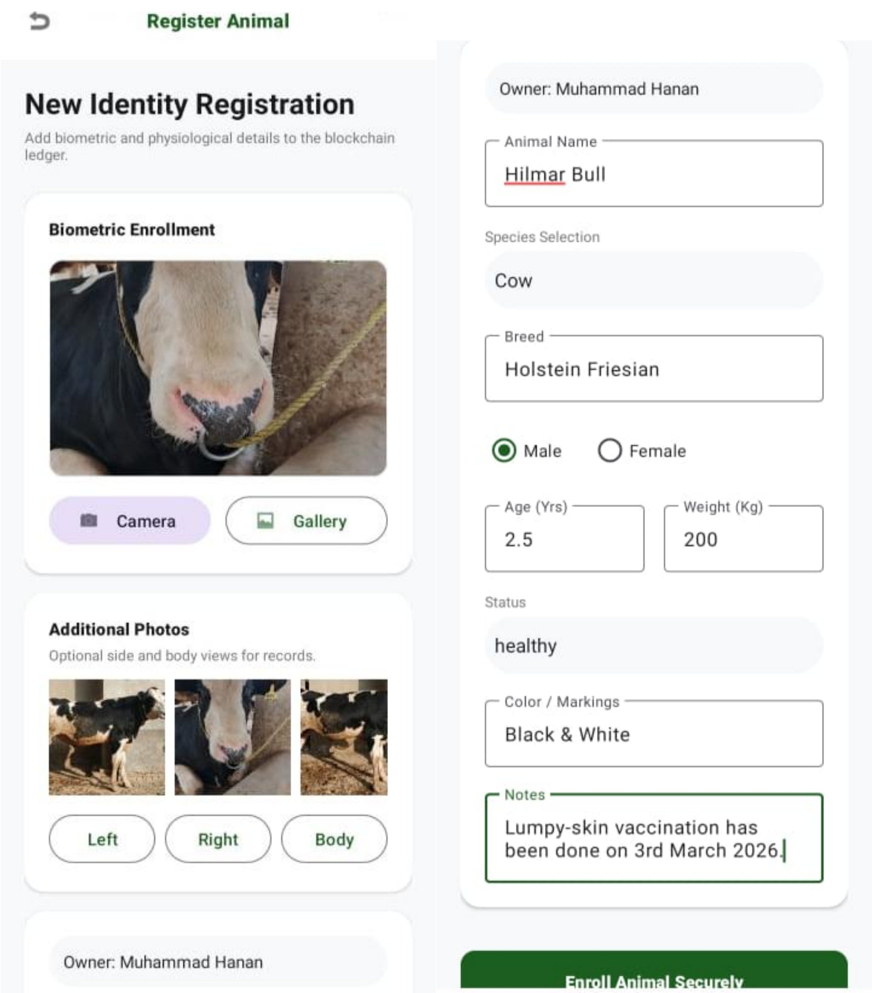
Users capture a muzzle image during registration.

Registration Process

1. Fill Animal Details
2. Capture Muzzle Image

3. AI Duplicate Check
4. Biometric Enrollment
5. Save Record

Screenshot



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11. Animal Identification

This feature identifies animals through muzzle recognition.

Process

1. Open Scan Module
2. Capture Animal Muzzle Image
3. Upload Image

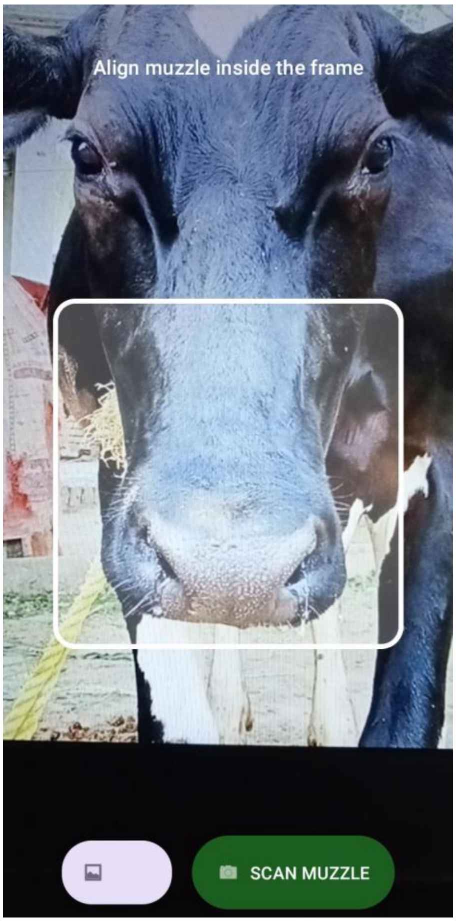
4. AI Matching Process
5. Display Match Results

Results

The system displays:

- Animal Name
- Owner Information
- Confidence Score
- Identification Status

Screenshot



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12. Animal Profile Management

The profile page stores complete livestock information.

Information Displayed

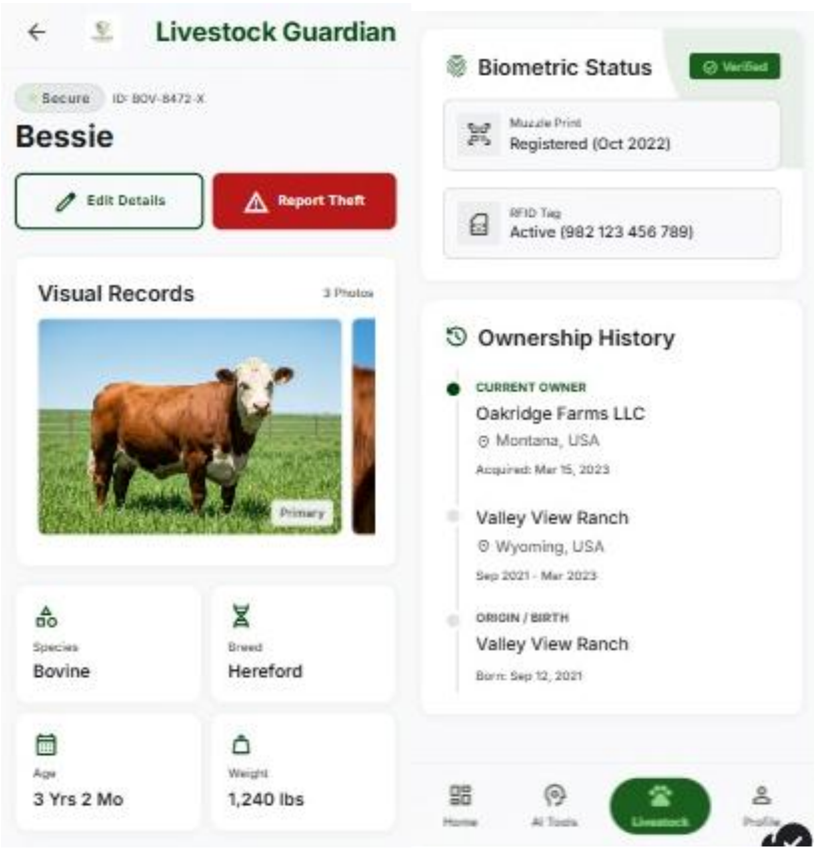
- Animal Name

- Species
- Breed
- Gender
- Age
- Weight
- Owner Details
- Registration Date
- Status

Available Actions

- View Profile
- Update Information
- View History
- Report Theft

Screenshot



13. Theft Reporting

The theft reporting system helps record stolen animals.

Reporting Steps

1. Open Animal Profile
2. Select Report Theft
3. Enter Incident Details
4. Enter Location
5. Submit Report

Stored Information

- Reporter Information
- Animal Information
- Location
- Description
- Timestamp

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14. Ownership Tracking

The application maintains ownership history records.

Tracked Information

- Previous Owner
- New Owner
- Transfer Date
- Transfer Reason
- Verification Status

This feature improves traceability and ownership verification.

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15. AI-Based Biometric Recognition

Livestock Guardian uses muzzle biometric recognition.

Why Muzzle Biometrics?

Each animal possesses a unique muzzle pattern similar to a human fingerprint.

AI Workflow

Capture Image

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Feature Extraction

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Embedding Generation

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Similarity Matching

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Identification Result

Benefits

- Difficult to Forge
- Accurate Identification
- Permanent Identity
- Theft Prevention

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16. Database Structure

The application uses seven major database tables.

Users

Stores user information.

Livestock

Stores animal information.

Livestock Images

Stores animal image references.

Embeddings

Stores biometric feature vectors.

Ownership History

Tracks ownership changes.

Theft Reports

Stores theft incidents.

AI Logs

Stores AI processing records.

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17. Security Features

Authentication

Secure user authentication system.

Data Protection

Encrypted communication between mobile app and server.

Cloud Security

Data stored securely using Supabase infrastructure.

API Security

Protected API endpoints and access control.

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18. Troubleshooting

Problem

Unable to Login

Solution

- Verify credentials
- Check internet connection
- Reset password if necessary

Problem

Image Upload Fails

Solution

- Check internet connection
- Verify camera permissions
- Retry image capture

Problem

Identification Fails

Solution

- Ensure animal is registered
- Capture clear muzzle image
- Verify server availability

Problem

Application Crashes

Solution

- Restart application
- Clear cache
- Reinstall application

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19. Frequently Asked Questions

Q1: Can multiple animals have the same biometric identity?

No. Each animal's muzzle pattern is unique.

Q2: Does the application work offline?

Internet connectivity is required for AI processing and cloud synchronization.

Q3: Is livestock data stored securely?

Yes. Data is stored using secure cloud infrastructure.

Q4: Can ownership history be viewed?

Yes. Ownership records are maintained for traceability.

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20. Future Enhancements

Future versions may include:

- Push Notifications
- Veterinary Health Records
- Marketplace Integration
- Offline Recognition Support
- Advanced Analytics Dashboard

- Multi-Language Support
- QR Code Integration
- Government Livestock Registry Integration

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21. Conclusion

Livestock Guardian provides an intelligent and secure livestock management solution through biometric identification, cloud computing, and artificial intelligence. The platform improves livestock traceability, reduces ownership disputes, enhances theft prevention, and supports digital transformation in the livestock sector.

Thank you for using Livestock Guardian.